PHASE EXTERNAL DESCRIPTION:

An Inter-Process Communication (ICP) allows tasks to communicate with each other using a message-passing. Every task is represented by a Task Control Block(TCB). Each TCB contains an id number, task name, state, and mailbox. For the ICP, the mailbox of the TCB uses a queue of message objects. The messages can be sent, received, counted, printed, and deleted between tasks.

The Scheduler, TCB, and IPC manage the task. The Scheduler uses a linked list of TCB and is responsible for task creation, scheduler, and retrieval. The TCB is a single task with ID, task name, state, and mailbox. There is a pointer at the head, and a pointer at the next task. The ICP makes communication between tasks. It interacts with the Scheduler to locate the destination of the task.

Passing a message uses Message_Send() and Message_Receive(). Message_Send() retrieves the destination TCB from the Scheduler and creates a message object then moves it to the mailbox queue. Message_Recieve() checks if the mailbox has messages, grabs the first message in FIFO order, removes it from the queue, and returns it to the caller.

The Message_Count(int task_id) function returns the number of messages in a specific task' mailbox, and Message_Count() gets the total number of from each task. The Message_DeleteAll(int task_id) clears the task's mailbox until all messages from its queue are gone. The Message_Print(int task_id) function copies the mailbox and displays the content of task's mailbox in such a way below:

Task Number: 1
Message Count: 1
Mail Box:
Source Task-id | Destination Task-id | Message Content | Message Size | Message Type | Message Arrival Time
3 | 1 | Reply from Task 3 to Task 1 | 27 | 1 | 1774972762

The ipc_Message_Dump() gives an overview, inerating through all possible task IDs and printing active task with message count of the task.